Reghunaath Ajith Kumar Ahila

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EDUCATION

Northeastern University, Boston, MA, USA 2024 – 2026
Masters of Science in Data Science | CGPA: 3.9
Vellore Institute of Technology, Vellore, TN, India 2018 – 2022
Bachelors of Technology in Computer Science and Engineering | CGPA: 8.56

WORK EXPERIENCE

QuantUniversity, Boston, MA Jul 2025 – Aug 2025
Graduate Intern
- Built a platform using **React** and **FastAPI** to enable **AI assisted educational content creation** and seamless hosting of generated materials, reducing content development time from 5 days to ~3 hours
- Identified and **fixed a critical paywall bypass vulnerability** in the first week, strengthening platform security
- Designed and implemented authentication and authorization systems addressing all the security and compliance for the **ISO** and **SOC2** certification

Infosys, Bengaluru, India Aug 2023 – Jul 2024
Senior Systems Engineer
- Developed and deployed a full-stack application with a **.NET microservice architecture** and **React.js** frontend, modernizing a legacy insurance platform through **REST APIs**, **JWT based authentication**, and **Redux** state management for a seamless user experience.
- Built a **.NET** rule based recommendation engine integrating 9 external systems through **gRPC, SOAP, and REST APIs**, with **SQL** used to cache external system data for efficient rule evaluation and policy recommendations.
- Developed a **Python script** to auto generate **unit test** cases and **Postman integration test cases** from business owned Excel sheets, reducing test case updates from hours to minutes per cycle and saving over 65 hours of manual effort long term.

Danske IT (subsidiary of Danske Bank acquired by Infosys in 2023), Bengaluru, India Jul 2022 – Aug 2023
Associate Software Engineer
- Developed and integrated **Camunda BPM workflows** within the **.NET** backend to orchestrate customer onboarding processes, automating task execution and service interactions to improve data processing efficiency, fault tolerance, and overall system reliability.
- Built and owned **CI/CD** pipelines on **Azure DevOps**, ensuring smooth deployment workflows, version control, and continuous integration across development and production environments.
- Integrated automated load testing with **Grafana K6** into the **CD** pipeline to evaluate system performance and ensure scalability under high traffic
- Independently implemented a monitoring solution using **Kibana (Elastic Stack)** to provide real time insights and enhance oversight across multiple team projects

Apprentice Jan 2022 – Jul 2022
- Gained comprehensive experience in fintech software development, working across **testing, DevOps, frontend, and backend** in an agile environment
- Improved **unit test** line coverage from 60% to 95% for **.NET backend**

PROJECTS

- **RescueLine AI** – Developed an AI powered emergency call triage system using **Twilio, ElevenLabs, and FastAPI** to automatically classify and route emergency calls by urgency during natural disasters, when traditional helplines are overwhelmed. Built a **real time voice AI agent** and **live dashboard** with **WebSocket** based updates for emergency coordinators managing high call volumes. Earned **1st place and $700 at Northeastern's Innovate 2026 Hackathon.**
- **LeadCatch AI** – Developed an AI chat assistant powered by **ChatGPT** and **Twilio APIs** to turn missed calls into booked appointments for small businesses. Designed a scalable Python backend for multi user handling and automated SMS based lead conversion. Earned **2nd place and $1500** in the "Best Startup Demonstrating Traction" category at the **Yconic AI Hackathon (Microsoft, Cambridge, MA)**.

TECHNICAL SKILLS

- **Backend:** C# (.NET), Java (Spring Boot), Python (Django, Flask, FastAPI)
- **Frontend:** React, Angular, Flutter
- **DevOps:** AWS, Azure DevOps, git
- **Other Programming Languages:** C, C++, MATLAB, R, Go
- **Testing:** JUnit, NUnit, xUnit, Postman, Selenium, k6
- **ML/AI:** TensorFlow, Keras, PyTorch, scikit-learn, Pandas, Numpy, OpenCV, Matplotlib, Seaborn, Plotly, LLMs, Streamlit, GenAI, LangChain, LangGraph, ElevenLabs
- **DBMS:** SQL, MongoDB

PUBLICATIONS

Muralidharan, K., Ramesh, A., Rithvik, G., Prem, S., Reghunaath, A. A., & Gopinath, M. P. (2021). 1D Convolution approach to human activity recognition using sensor data and comparison with machine learning algorithms. *International Journal of Cognitive Computing in Engineering*, *2*, 130-143. (63 citations)